

Linear Algebra Axler - Solutions Manual

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Chapter 1

Vector Spaces

1.1 $\mathbb{R}^n$ and $\mathbb{C}^n$

Exercise 1.A.1

$$(a + bi)(c + di) = 1$$
$$\Rightarrow ac - bd = 1 \text{ and } ad + bc = 0$$

Solving for c and d gives

$$c = \frac{a}{a^2 + b^2} \text{ and } d = \frac{-b}{a^2 + b^2}$$

Hence,

$$\frac{1}{a + bi} = \frac{a}{a^2 + b^2} - \frac{b}{a^2 + b^2}i$$

Exercise 1.A.2

Just compute it.

$$\left(\frac{-1 + \sqrt{3}i}{2} \right)^3 = 1$$

Exercise 1.A.3

$$(a + bi)^2 = i$$
$$\Rightarrow a^2 - b^2 = 0 \text{ and } 2ab = 1$$
$$\Rightarrow |a| = |b| \text{ and } ab = \frac{1}{2}$$

Hence, the roots are

$$-\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i \text{ and } \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$$

Exercise 1.A.4

$$\begin{aligned}
& \alpha + \beta \\
&= (a + bi) + (c + di) \\
&= (a + c) + (b + d)i \\
&= (c + a) + (d + b)i \\
&= (c + di) + (a + bi) \\
&= \beta + \alpha
\end{aligned}$$

Exercise 1.A.5

Too lazy.

Exercise 1.A.6

Too lazy.

Exercise 1.A.7

Given $\alpha = a + bi$, choose $\beta = -\alpha = -a - bi$.

$$\alpha + \beta = (a - a) + (b - b)i = 0$$

Exercise 1.A.8

Given $\alpha = a + bi$, choose $\beta = \frac{\bar{\alpha}}{|\alpha|^2} = \frac{a-bi}{a^2+b^2}$.

$$\alpha\beta = \frac{(a+bi)(a-bi)}{a^2+b^2} = \frac{a^2+b^2}{a^2+b^2} = 1$$

Exercise 1.A.9

Too lazy.

Exercise 1.A.10

$$\begin{aligned}
2x &= (5, 9, -6, 8) - (4, -3, 1, 7) = (1, 12, -7, 1) \\
&\Rightarrow x = \left(\frac{1}{2}, 6, -\frac{7}{2}, \frac{1}{2}\right)
\end{aligned}$$

Exercise 1.A.11

Vectors not collinear.

Exercise 1.A.12

Too lazy.

Exercise 1.A.13

Too lazy.

Exercise 1.A.14

$$1x = 1(x_1, x_2, \dots, x_n) = (1x_1, 1x_2, \dots, 1x_n) = (x_1, x_2, \dots, x_n) = x$$

Exercise 1.A.15

Too lazy.

Exercise 1.A.16

Too lazy.

1.2 Definition of Vector Space

Exercise 1.B.1

We use commutativity.

$$(-v) + v = v + (-v) = 0$$

$$\Rightarrow -(-v) = v$$

Exercise 1.B.2

Suppose for contradiction that $a \neq 0$ and $v \neq 0$. Then $v_i \neq 0$ for some i . Hence, $av = (av_1, \dots, av_i, \dots, av_n)$ where $av_i \neq 0$, which contradicts $av = 0$.

Exercise 1.B.3

$v + (-v + w) = (v + (-v)) + w = 0 + w = w$, hence we can set $3x = -v + w$, so $x = \frac{1}{3}(-v + w)$ which clearly exists by the axioms.

Exercise 1.B.4

The identity is missing.

Exercise 1.B.5

For every v , we can create $(-1)v$. Clearly, $(-1)v \in V$. Also,

$$v + (-1)v = 1v + (-1)v = (1 + (-1))v = 0v = 0$$

where we used $0v = 0$ in the last equality. Hence, every $v \in V$ has an inverse $(-1)v$.

Exercise 1.B.6

No, it's not associative:

$$(\infty + (-\infty)) + (-\infty) = 0 + (-\infty) = -\infty$$

but

$$\infty + ((-\infty) + (-\infty)) = \infty + (-\infty) = 0$$

1.3 Subspaces**Exercise 1.C.1**

(a) Yes. $(0, 0, 0)$ satisfies the condition. If $x_1 + 2x_2 + 3x_3 = 0$ and $y_1 + 2y_2 + 3y_3 = 0$, then $(x_1 + y_1) + 2(x_2 + y_2) + 3(x_3 + y_3) = 0$, hence it's closed under addition. Also, if $x_1 + 2x_2 + 3x_3 = 0$ and $a \in \mathbb{F}$, then $ax_1 + 2ax_2 + 3ax_3 = 0$, hence it's closed under scalar multiplication.

(b) No. $0 \notin V$.

(c) No. It's not closed under addition. $x_1x_2x_3 = 0$ implies at least one of them is 0. For example, we have $(1, 0, 0)$ and $(0, 1, 1)$. But their sum $(1, 1, 1)$ does not satisfy the conditions.

(d) Yes. $(0, 0, 0)$ satisfies the condition. If $x_1 = 5x_3$ and $y_1 = 5y_3$, then $x_1 + y_1 = 5x_3 + 5y_3 = 5(x_3 + y_3)$, so it's closed under addition. If $x_1 = 5x_3$ and $a \in \mathbb{F}$, then $ax_1 = 5ax_3$ still holds, so it's closed under scalar multiplication.

Exercise 1.C.2

Whut is dat. Too lazy.

Exercise 1.C.3

The constant zero function $f_0 : (-4, 4) \rightarrow \mathbb{R}$ with $f_0(x) = 0$ for all $x \in (-4, 4)$ is the additive identity. Clearly, $f'_0(-1) = 0 = 3f_0(2)$.

Now, for any two functions f, g satisfying the conditions, i.e. $f'(-1) = 3f(2)$ and $g'(-1) = 3g(2)$, and denoting $(f + g)(x) = f(x) + g(x)$, we have $(f + g)'(-1) = f'(-1) + g'(-1) = 3f(2) + 3g(2) = 3(f + g)(2)$, where the first equality comes from the linearity of differentiation.

Finally, if $f'(-1) = 3f(2)$ and $a \in \mathbb{F}$, and denoting $(af)(x) = af(x)$, then $(af)'(-1) = af'(-1) = a(3f(2)) = 3(af)(2)$, where the first equality comes from the linearity of differentiation.

Exercise 1.C.4

If $\int_0^1 f = \int_0^1 g = b$, then $\int_0^1 (f + g) = 2b$. We want the vector space to be closed under addition, so the sum must also satisfy the condition, namely that its integral be b . Hence, we require $2b = b$, and 0 is the only solution.

Likewise, if the integrals of functions are zero, then the integral of their sum is also zero, by linearity of integration. The zero function, which is the additive identity, satisfies the condition as well. Finally, integration's linearity also guarantees that the scaled function also has integral zero.

Exercise 1.C.5

Depends on the field. If the field is limited to real numbers, then yes, it's a subspace. If the field is the entire complex numbers, then obviously it's not closed under scalar multiplication, so not a subspace.

Exercise 1.C.6

(a) Yes. $a^3 = b^3$ implies $a = b$. The additive identity $(0, 0, 0)$ clearly satisfies the condition. If $x_1 = x_3$ and $y_1 = y_3$, then $x_1 + y_1 = x_3 + y_3$, so it's closed under addition. And if $x_1 = x_3$ and $a \in \mathbb{F}$, then $ax_1 = ax_3$ still holds.

(b) No. In the complex numbers, $a^3 = b^3$ can potentially admit 3 solutions.

Exercise 1.C.7

Take all vectors of the form (a, b) where $a, b \in \mathbb{Z}$. Since the field is still $\mathbb{R}$, then we are missing the space between the lattice points, so it's not closed under scalar multiplication.

Exercise 1.C.8

Take all vectors of the form $(a, 0)$ and $(0, b)$ where $a, b \in \mathbb{R}$. It's not closed under addition.

Exercise 1.C.9

No. It's not closed under addition. When the periods of two functions are incommensurable, their sum is aperiodic. For example, let

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Z} \\ 0 & \text{otherwise} \end{cases}$$

$$g(x) = \begin{cases} 1 & \text{if } x = k\pi \text{ for } k \in \mathbb{Z} \\ 0 & \text{otherwise} \end{cases}$$

In other words, f evaluates to 1 at every integer, while g evaluates to 1 at every multiple of π . Let $h = f + g$. Clearly, $h(0) = 2$. However, it will never reappear,

because $f(x) = 1$ only at rational x , whereas $g(x) = 1$ only at irrational x . Thus $h(0) \neq h(0 + p)$ for all $p \in \mathbb{R}$. Hence h is aperiodic.

Exercise 1.C.10

Since $0 \in U_1$ and $0 \in U_2$, then $0 \in U_1 \cap U_2$.

If $v_1, v_2 \in U_1 \cap U_2$, then $v_1, v_2 \in U_1$, so $v_1 + v_2 \in U_1$. Likewise, the intersection also gives us $v_1, v_2 \in U_2$, so $v_1 + v_2 \in U_2$. Hence $v_1 + v_2 \in U_1 \cap U_2$.

Finally, if $v \in U_1 \cap U_2$, then $v \in U_1$, so $av \in U_1$. Likewise, we also have $v \in U_2$, so $av \in U_2$. Hence $av \in U_1 \cap U_2$.

Exercise 1.C.11

Simply extend Exercise 1.C.11 to the infinite case.

Exercise 1.C.12

Let $U_1 \cup U_2$ be a subspace. Suppose that $U_1 \not\subseteq U_2$ and $U_2 \not\subseteq U_1$. Then I can find $v_1 \in U_1 \setminus U_2$ and $v_2 \in U_2 \setminus U_1$. Then $v_1 + v_2 = v_3 \notin U_1$, because otherwise $v_3 - v_1 = v_2 \in U_1$, contradicting $v_2 \in U_2 \setminus U_1$. Likewise, $v_3 \notin U_2$. Thus, it's not closed under addition, so $U_1 \cup U_2$ would not be a subspace. Hence, one of $U_1 \subseteq U_2$ and $U_2 \subseteq U_1$ must be true.

The other direction is easy.

Exercise 1.C.13

Too lazy. Maybe hard, idk.

Exercise 1.C.14

Too lazy.

Exercise 1.C.15

$U + U = U$, because it's closed under addition.

Exercise 1.C.16

Well, $U + W$ is a subspace, hence its addition is commutative, so indeed $U + W = W + U$.

Exercise 1.C.17

Again, yes. $U_1 + U_2 + U_3$ is a subspace, so its addition is associative, so indeed $(U_1 + U_2) + U_3 = U_1 + (U_2 + U_3)$.

Exercise 1.C.18

There is an additive identity, which is the space $\{0\}$. However, the only space that has an additive inverse is $\{0\}$ itself. Every other space contains some form of $\{0, v, \dots\}$. If we try to find an inverse space, it will also contain some form of $\{0, w, \dots\}$. Taking their sum, we realize we can always select 0 from the first space, and some nonzero element w from the second space, to give $0 + w = w$. Hence, their sum can never be $\{0\}$.

Exercise 1.C.19

False. Let $W = \{(a, 0) \mid a \in \mathbb{R}\}$, which is visually a horizontal line passing through the origin. Let $U_1 = \{(a, a) \mid a \in \mathbb{R}\}$, which is visually a diagonal line passing through the origin. Let $U_2 = \{(0, a) \mid a \in \mathbb{R}\}$, which is visually a vertical line passing through the origin. Clearly, $U_1 + W$ and $U_2 + W$ both give the whole space $\mathbb{R}^2$. But $U_1 \neq U_2$.

Exercises 1.C.20+

Too lazy.

Chapter 2

Finite-Dimensional Vector Spaces

2.1 Span and Linear Independence

Exercise 2.A.1

Compute their linear combination:

$$\begin{aligned} & a_1(v_1 - v_2) + a_2(v_2 - v_3) + a_3(v_3 - v_4) + a_4(v_4) \\ &= a_1v_1 + (a_2 - a_1)v_2 + (a_3 - a_2)v_3 + (a_4 - a_3)v_4 \end{aligned}$$

Since v_1, v_2, v_3, v_4 span V , then every $v \in V$ can be expressed

$$v = b_1v_1 + b_2v_2 + b_3v_3 + b_4v_4$$

But we can also set $a_1 = b_1$ and $a_2 = b_2 + a_1$ and $a_3 = b_3 + a_2$ and $a_4 = b_4 + a_3$. Hence, they span V as well.

Exercise 2.A.2

Too lazy.

Exercise 2.A.3

Simply find t such that the third vector is not contained in the span of the first two vectors.

Exercise 2.A.4

Too lazy.

Exercise 2.A.5

(a) If the field is real numbers, then the real part and complex part behave like independent components of 2D vectors, namely $(1, 1)$ and $(1, -1)$. They are clearly linearly independent.

(b) With the complex numbers at our disposal, we can find $z \in \mathbb{C}$ such that $z(1 + i) = 1 - i$. Hence one is in the span of the other.

Exercise 2.A.6

We want

$$a_1(v_1 - v_2) + a_2(v_2 - v_3) + a_3(v_3 - v_4) + a_4v_4 = 0$$

which can be rewritten

$$a_1v_1 + (a_2 - a_1)v_2 + (a_3 - a_2)v_3 + (a_4 - a_3)v_4 = 0$$

Since v_1, v_2, v_3, v_4 are linearly independent, then we must have $a_1 = 0$ and $a_2 - a_1 = 0$ and $a_3 - a_2 = 0$ and $a_4 - a_3 = 0$. It's easy to see that these equations imply every $a_i = 0$.

Exercise 2.A.7

Similarly to the previous exercise, we have

$$5a_1v_1 + (a_2 - 4a_1)v_2 + a_3v_3 + \cdots + a_mv_m = 0$$

Hence

$$5a_1 = 0$$

$$\Rightarrow a_1 = 0$$

$$\Rightarrow a_2 = 0$$

So yes, they are linearly independent.

Exercise 2.A.8

We simply factor out:

$$a_1(\lambda v_1) + \cdots + a_m(\lambda v_m) = 0$$

$$\lambda(a_1v_1 + \cdots + a_mv_m) = 0$$

$$\Rightarrow a_1v_1 + \cdots + a_mv_m = 0$$

So yes, they are linearly independent.

Exercise 2.A.9

False. Just take $v_1 = w_2 = (1, 0)$ and $v_2 = w_1 = (0, 1)$. The sums are $v_1 + w_1 = (1, 1)$ and $v_2 + w_2 = (1, 1)$, which are clearly not linearly independent since they are identical.

Exercises 2.A.10+

Too lazy.

2.2 Bases**Exercise 2.B.1**

$\{0\}$ obviously has a single basis. Also, any space of the form $\{0, v\}$ with field $\{0, 1\}$ has a single basis $\{v\}$. For any other bigger field $\{0, 1, a, \dots\}$, we would have $av \neq v$ so it would be an additional element which forms a new basis. For any bigger space $\{0, v, w\}$, if they are linearly independent, then both $\{v, w\}$ and $\{v + w, w\}$ are bases. If they are linearly dependent, then clearly both $\{v\}$ and $\{w\}$ can single-handedly become the basis.

Exercise 2.B.2

Too lazy.

Exercise 2.B.3

(a)

$$\{(3, 1, 0, 0, 0); (0, 0, 7, 1, 0); (0, 0, 0, 0, 1)\}$$

(b) Append the following vectors:

$$(1, 0, 0, 0, 0); (0, 0, 1, 0, 0)$$

(c) Using the result from (b):

$$W = \{(a, 0, b, 0, 0) \mid a, b \in \mathbb{R}\}$$

Exercise 2.B.4

Too lazy.

Exercise 2.B.5

True. Each polynomial $\sum_k^n a_k x^k$ is akin to a vector $(a_1, \dots, a_n)$ in the polynomial vector space. Clearly, there exists bases where none of the individual component entries are zero.

Exercise 2.B.6

Too lazy. Clearly, it's linearly independent. You can use the concept of dimension from the next chapter to answer the spanning question.

Exercise 2.B.7+

Too lazy.

2.3 Dimension**Exercise 2.C.1**

If U is a subspace of V , then there exists W such that $U \oplus W = V$. So we have

$$\dim U + \dim W - \dim(U \cap W) = \dim V$$

Since $U \cap W = \{0\}$, then $\dim(U \cap W) = 0$. With the fact that $\dim U = \dim V$, we cancel both sides and are left with

$$\dim W = 0$$

So $W = \{0\}$. Hence,

$$U \oplus \{0\} = V$$

which implies $U = V$.

Exercise 2.C.2

Every subspace must contain the additive identity $0 = (0, 0)$. Hence, the latter forms the trivial subspace $\{0\}$.

If we include an additional vector in a subspace U , i.e. $\{0, v\} \subseteq U$, then $av \in U$ as well, for every $a \in \mathbb{R}$. These represent lines in $\mathbb{R}^2$.

If we include two vectors, i.e. $\{0, v, w\} \subseteq U$, then either $w \in \text{span}\{v\}$ or not. In the former case, we have $w = av$ so we are still left with a line. In the latter case, v, w are linearly independent, so U has dimension 2. Using the result from the previous exercise, it follows that $U = \mathbb{R}^2$.

Exercises 2.C.3 - 2.C.10

Too lazy.

Exercise 2.C.11

$$\dim \mathbb{R}^8 = \dim U + \dim W - \dim(U \cap W)$$

$$8 = 3 + 5 - \dim(U \cap W)$$

$$\Rightarrow \dim(U \cap W) = 0$$

$$\Rightarrow U \cap W = \{0\}$$

Exercise 2.C.12

$$\begin{aligned}
\dim \mathbb{R}^9 &\geq \dim(U + W) = \dim U + \dim W - \dim(U \cap W) \\
9 &\geq 5 + 5 - \dim(U \cap W) \\
&\Rightarrow \dim(U \cap W) \geq 1
\end{aligned}$$

Since $\dim\{0\} = 0$, then it must follow that

$$\Rightarrow U \cap W \neq \{0\}$$

Exercise 2.C.13

$$\begin{aligned}
\dim \mathbb{C}^6 &\geq \dim(U + W) = \dim U + \dim W - \dim(U \cap W) \\
6 &\geq 4 + 4 - \dim(U \cap W) \\
&\Rightarrow \dim(U \cap W) \geq 2
\end{aligned}$$

Hence, there exists two basis vectors. Clearly, none is in the span of the other, which means none is a scalar multiple of the other.

Exercise 2.C.14

$$\dim(U_1 + U_2) = \dim U_1 + \dim U_2 - \dim(U_1 \cap U_2)$$

Since dimension is always non-negative, it follows that

$$\Rightarrow \dim(U_1 + U_2) \leq \dim U_1 + \dim U_2$$

Simply apply this rule in an induction fashion.

Exercises 2.C.15+

Too lazy.

Chapter 3

Linear Maps

3.1 The Vector Space of Linear Maps

Exercise 3.A.1

Zero must be mapped to zero, hence for the first component: $2(0) - 4(0) + 3(0) + b = 0$. So $b = 0$. Linear maps must obey the scalar multiple rule. So the second component must satisfy $T(\lambda x, \lambda y, \lambda z) = \lambda T(x, y, z)$, hence

$$\begin{aligned} 6(\lambda x) + c(\lambda x)(\lambda y)(\lambda z) &= \lambda(6x + cxyz) \\ \Rightarrow \lambda^3 cxyz &= \lambda cxyz \end{aligned}$$

The only way for this equality to hold for all λ, x, y, z is by setting $c = 0$.

Exercise 3.A.2

Too lazy. Just the linear properties of polynomials and differentiation and integration.

Exercise 3.A.3

These form a basis for F^n , called the trivial basis:

$$\begin{aligned} e_1 &= (1, 0, 0, \dots, 0) \\ e_2 &= (0, 1, 0, \dots, 0) \\ &\vdots \\ e_{n-1} &= (0, 0, \dots, 1, 0) \\ e_n &= (\underbrace{0, 0, \dots, 0}_n, 1) \end{aligned}$$

When applying T on a vector x , we represent it using the trivial basis e_i , and we employ linearity as follows:

$$\begin{aligned} T(x) &= T(x_1, x_2, \dots, x_n) \\ &= T(x_1 e_1 + x_2 e_2 + \dots + x_n e_n) \\ &= x_1 T(e_1) + x_2 T(e_2) + \dots + x_n T(e_n) \end{aligned}$$

Notice that $T(e_i)$ obviously do not depend on the input x . So they are constants inherent to the linear map. They are also vectors which *exist* in F^m . We can enumerate their components:

$$\begin{aligned} T(e_1) &= (A_{11}, A_{21}, \dots, A_{m1}) \\ T(e_2) &= (A_{12}, A_{22}, \dots, A_{m2}) \\ &\vdots \\ T(e_n) &= (A_{1n}, A_{2n}, \dots, A_{mn}) \end{aligned}$$

We can use this representation, plugging it back into the previous set of equation

$$\begin{aligned} T(x) &= x_1 (A_{11}, A_{21}, \dots, A_{m1}) + \dots + x_n (A_{1n}, A_{2n}, \dots, A_{mn}) \\ &= (A_{11}x_1 + A_{12}x_2 + \dots + A_{1n}x_n, \quad \dots, \quad A_{m1}x_1 + A_{m2}x_2 + \dots + A_{mn}x_n) \end{aligned}$$

To reiterate, every vector x follows this transformation, because they can all be represented in the trivial basis, the latter which transform identically each time.

Exercise 3.A.4

Suppose

$$a_1 v_1 + \dots + a_m v_m = 0$$

Applying T on both sides and leveraging linearity, we get

$$\begin{aligned} T(a_1 v_1 + \dots + a_m v_m) &= T(0) \\ \Rightarrow a_1 T v_1 + \dots + a_m T v_m &= 0 \end{aligned}$$

where we used the fact that $T(0) = 0$. Since they are linearly independent, then each of $a_1, \dots, a_m$ also equals zero.

Exercises 3.A.5 - 3.A.6

Too lazy.

Exercise 3.A.7

If $\dim V = 1$, then it has a basis of length 1, let's say $\{e\}$. Now,

$$T(e) = f$$

Since $f \in V$, then it's a linear combination of the basis, so $f = \lambda e$ for some $\lambda \in \mathbb{F}$. Well, λ is precisely our scaling factor. For every $v \in V$,

$$T(v) = T(ae) = aT(e) = a\lambda e = \lambda(ae) = \lambda v$$

Exercise 3.A.8

$$\phi(x, y) = \sqrt{x^2 + y^2}$$

Points are mapped to their distance from the origin. In other words, points lying on the same circle (centered at the origin) are mapped to the same number, namely the radius of that circle. It satisfies the desired property:

$$\phi(ax, ay) = \sqrt{(ax)^2 + (ay)^2} = \sqrt{a^2(x^2 + y^2)} = a\sqrt{x^2 + y^2} = a\phi(x, y)$$

However, clearly, it doesn't satisfy the addition property.

Exercise 3.A.9

$$\phi(z) = \bar{z}$$

It satisfies the desired property:

$$\phi(x + y) = \overline{x + y} = \bar{x} + \bar{y} = \phi(x) + \phi(y)$$

However, it fails the complex scalar multiplication:

$$\begin{aligned}\phi(i(a + bi)) &= \overline{i(a + bi)} = \overline{ai - b} = -ai - b \\ &\neq i(\phi(a + bi)) = i(\overline{a + bi}) = i(a - bi) = ai + b\end{aligned}$$

I found this online.

Exercise 3.A.10

We can always extend a basis e_i of U into a basis e_i, f_i of V . Suppose $v \notin U$. Then

$$T(v) = 0$$

However, we can represent it using the extended basis. Assuming that T is linear, we have

$$\begin{aligned}T(v) &= T(a_1e_1 + \cdots + a_me_m + b_1f_1 + \cdots + b_nf_n) \\ &= T(u + w)\end{aligned}$$

for some $u \in U$ and $w \notin U$.

$$\begin{aligned} &= T(u) + T(w) \\ &= T(u) \neq 0 \end{aligned}$$

This last inequality comes from the fact that since $S \neq 0$, then for some $u \in U$, we must have $T(u) = S(u) \neq 0$. In other words, they cannot all be zero, otherwise $T(v) = 0$ for all v . We are working with v 's that exhibit such property.

Anyways, this contradiction implies that T cannot be linear.

Exercise 3.A.11

Using the same idea as Exercise 3.A.10, we represent $v \in V$ in the extended basis. Simply let

$$T(u) = a_1 e_1 + \cdots + a_m e_m + 0f_1 + \cdots + 0f_n$$

where e_i are basis for U and f_1 are the extensions. In other words, set the coefficients to zero for all basis not contained in U . Obviously, $T(u) = S(u)$. Also, zero is mapped to zero. Additivity is satisfied. Scalar multiplication is satisfied. Hence, it's linear.

Exercise 3.A.12

Let V have dimension n , and consider a basis $e_1, e_2, \dots, e_n$. Consider the set of linear functions that map this basis to a portion of the basis $f_1, f_2, f_3, \dots$ of W :

$$\begin{aligned} T_1(e_i) &= f_i \\ T_2(e_i) &= f_{n+i} \\ T_3(e_i) &= f_{2n+i} \\ &\vdots \end{aligned}$$

These are well-defined functions. (In effect, the basis transformation uniquely determine every other vectors' transformation.)

Since each T_k affects a set of components in W that do not overlap with each other, then if we want the linear combination of these functions to produce the zero function, i.e.

$$\sum_{k=0}^{\infty} a_k T_k(v) = 0 \quad \text{for all } v \in V$$

then each a_k must be zero. Hence, these functions are linearly independent. This list of linearly independent functions can be extended to a basis of W . Since the list of functions is infinite, then the basis, which is its superset, must be infinite as well. Hence, $\mathcal{L}(V, W)$ is infinite-dimensional.

Exercise 3.A.13

??

Exercise 3.A.14

For every space V , we can find maps

$$S(x_1e_1 + x_2e_2) = (x_2e_1 + x_1e_2)$$

$$T(x_1e_1 + x_2e_2) = (-x_2e_1 + x_1e_2)$$

We compute

$$S(T(x_1e_1 + x_2e_2)) = S(-x_2e_1 + x_1e_2) = x_1e_1 - x_2e_2$$

$$T(S(x_1e_1 + x_2e_2)) = T(x_2e_1 + x_1e_2) = -x_1e_1 + x_2e_2$$

They are clearly different.

3.2 Null Spaces and Ranges**Exercise 3.B.1**

$$T : \mathbb{R}^5 \rightarrow \mathbb{R}^2$$

defined by

$$T(x_1, x_2, x_3, x_4, x_5) = (x_1, x_2)$$

Exercise 3.B.2

We rewrite

$$\text{range } S \subset \text{null } T$$

into

$$S(V) \subset \text{null } T$$

We derive. Since $T(V) \subseteq V$, we have

$$\Rightarrow S(T(V)) \subset \text{null } T$$

$$\Rightarrow T(S(T(V))) \subset T(\text{null } T)$$

$$\Rightarrow T(S(T(V))) \subset \{0\}$$

$$\Rightarrow S(T(S(T(V)))) \subset S(\{0\})$$

Since S is linear, it maps zero to zero:

$$\Rightarrow S(T(S(T(V)))) \subset \{0\}$$

Since linear maps are non-empty, then it must map to the singular zero:

$$\Rightarrow S(T(S(T(V)))) = \{0\}$$

$$\Rightarrow (ST)^2 = 0$$

Exercise 3.B.3

- (a) Surjectivity.
- (b) Injectivity.

Exercise 3.B.4

Not closed under addition.

$$T_1(x_1, x_2, x_3, x_4, x_5) = (x_1, x_2, 0, 0, 0)$$

$$T_2(x_1, x_2, x_3, x_4, x_5) = (0, 0, 0, x_4, x_5)$$

Each has $\dim \text{null} = 3 > 2$. But

$$(T_1 + T_2)(x_1, x_2, x_3, x_4, x_5) = (x_1, x_2, 0, x_4, x_5)$$

which has $\dim \text{null} = 1 \leq 2$.

Exercise 3.B.5

$$T(x_1, x_2, x_3, x_4) = (0, 0, x_1, x_2)$$

Exercise 3.B.6+

Too lazy.

3.3 Matrices**Exercise 3.C.1**

Given a map T , we can use its matrix representation:

$$T(v_1) = A_{11}w_1 + \cdots + A_{n1}w_n$$

$$\vdots$$

$$T(v_m) = A_{1m}w_1 + \cdots + A_{nm}w_n$$

where $w_1, \dots, w_n$ is a basis of $\text{range } T$. We argue that for each w_i , one of A_{ik} for some k must be nonzero.

Otherwise, suppose for contradiction that none of the basis vectors v_j produce a nonzero multiple of w_i . Then, none of the linear combinations of v_j produce a nonzero multiple of w_i . Which means w_i is not in the range of T , which would contradict the assumption that w_i is a basis vector of T .

Since we now know that there exist nonzero A_{ik} for each basis w_i , then the number of nonzero entries for the matrix A is at least $n = \dim \text{range } T$.

Exercise 3.C.2

Basis for $\mathcal{P}_3(\mathbb{R})$:

$$x^3 \quad x^2 \quad x \quad 1$$

Basis for $\mathcal{P}_2(\mathbb{R})$:

$$3x^2 \quad 2x \quad 1$$

Exercise 3.C.3

For every linear map, given a basis $e_1, \dots, e_n$ of V , we compute their image:

$$T(e_j) = f_j$$

We know that $f_1, \dots, f_n$ spans range T . We can reduce it to a basis by removing vectors. Using a similar approach to the textbook, when we find a vector contained in the span of the remaining vectors, we remove that vector. However, most likely, we will be removing a vector somewhere in the middle of $f_1, \dots, f_n$.

Let's say this happens at index k . When this happens, we relocate k to the end, and shift everything else. In other words, e_i and f_i remain unchanged for $i < k$, but $e_i, f_i \rightarrow e_{i-1}, f_{i-1}$ for $i > k$, and $e_k, f_k \rightarrow e_n, f_n$. All of this is possible simply because a basis is unordered.

Since we are working in finite-dimensional space, this procedure will eventually end. When it ends, we get a basis $f_1, \dots, f_m$. Looking back at our linear map, we notice that

$$T(e_j) = A_{jj}f_j$$

where re-ordering has been done, such that $A_{jj} = 1$ only for $j \leq m$, where $m = \dim \text{range } T$.

Exercise 3.C.4

See previous exercise. Predicted, I'm too genius.

Exercise 3.C.5

Too lazy.

Exercise 3.C.6

If every entry of $\mathcal{M}(T)$ is 1, then for any choice of basis for W , we have

$$T(e_k) = f_1 + \dots + f_n$$

for every k , where $n = \dim W$. Since every basis e_k is mapped to this sum, then every linear combination of them is mapped to this sum. Hence, this sum is a basis for range T , so its dimension is 1.

To prove the other direction, since $\dim \text{range } T = 1$, then it has a basis f such that $T(e_k) = a_k f$ for some $a_k \in \mathbb{F}$. In other words, all basis vectors of V

are mapped to a multiple of f . We can even do better: simply use $e'_k = \frac{1}{a_k}e_k$, which produces $T(e'_k) = T(\frac{1}{a_k}e_k) = f$. In other words, we can find a basis of V that are all mapped to f under T . We use these e'_k as our desired basis for V .

Now, we want to express f in a basis of W where each coefficient is 1:

$$f = f_1 + \cdots + f_n$$

Clearly, it's possible. Simply fix every $f_1, \dots, f_{n-1}$. Then solve for f_n :

$$f_n = f - (f_1 + \cdots + f_{n-1})$$

Hence, substituting into the first equation, we get

$$T(e'_k) = f = f_1 + \cdots + f_n$$

As you see, the entries of A are 1.

Exercises 3.C.7+

Too lazy.

3.4 Invertibility and Isomorphic Vector Spaces